

GROUP 0 – THE NOBLE GASES

KEY VOCABULARY

Noble gas: a Group 0 element with a full outer shell.

Inert: very unreactive.

Full outer shell: a stable arrangement of electrons.

Boiling point: increases going down Group 0.

COMMON MISCONCEPTIONS

~~— Noble gases are reactive like other groups.~~

✓ They are very unreactive (inert).

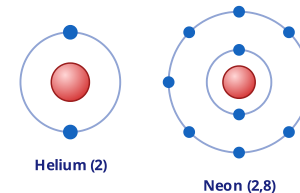
~~— Helium has 8 outer electrons.~~

✓ Helium has 2 — but its shell is still full.

~~— Boiling points fall down the group.~~

✓ Boiling points rise down Group 0.

FULL OUTER SHELLS



A **full outer shell** means no need to gain, lose or share electrons — so they are **unreactive**.

1 WHY UNREACTIVE

Explain, in terms of electrons, why noble gases are unreactive. [2 marks]

2 STATE AT RTP

State the physical state of the noble gases at room temperature. [1 mark]

3 BOILING POINT TREND

Describe and explain the trend in boiling point down Group 0. [2 marks]

4 PREDICT A VALUE

Argon boils at -186°C , krypton at -153°C . Predict a boiling point for neon. [1 mark]

Answer: _____ $^{\circ}\text{C}$

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Noble gases have a ____ outer shell, which makes them ____.

full

unreactive

empty

reactive

6 USES

Suggest why helium is used in balloons and argon in light bulbs. [2 marks]
